

Definition:

- IOT is the system of physical devices & things that can be connected to each other through the help of internet for sharing of data.

Characteristics (How it works)

- Sensors
- Connectivity of internet
- low power consumption
- Cloud storage
- Communication devices
- Remote access
- Security

Advantages

- Save time
- More Safety
- Health Monitoring.
- Smart city

There are two ways of building IOT.

- 1) From a separate networking including only physical objects.
- 2) Make the internet ever more expansive, but this requires hard-core technologies such as rigorous cloud computing and rapid big data storage.

IOT advantages =

The advantages of IOT span across every area of lifestyle and business. Here is a list of some of the advantages that IOT has to offer.

• Improved customer Engagement:

- current analytics suffers from blind-spots and significant flaws in accuracy and as noted, engagement remains passive.

• Technology optimization:

- The same technologies and data which improve the customer experience also known to improve device use.

Reduced waste

- IOT makes areas of improvement clear.

Enhanced Data Collection:

- Modern data collection suffers from its limitations and its design for passive use.

IOT devices and Embedded Devices

- An embedded device is a small computer that is ~~fixed~~ used inside a machine that can do one task. It cannot connect to the internet and work by itself.
- An IOT device is a smart device that can connect to internet, using sensors it sense the data and sends or receives the data through internet. It can perform multiple tasks.

Difference between IOT device and Embedded devices

- | IOT devices | Embedded devices |
|---|---------------------------------------|
| - Is a smart device that | - A device that do a fixed |
| - can connect to internet and share data. | - job using a small computer inside. |
| - It can connect to internet. | - It cannot connect to internet. |
| - Smart control through the help of mobile phone. | - Does not support any smart control. |
| - can get updates online. | - cannot get's updates. |

Ex: Smartwatch, smart city, motion sensors with data detection / collection

Eg: washing machines, digital watches, calculators, microwave ovens.

Properties of IOT devices:-

- i) Connectivity
- ii) Sensors, Actuators: works on the sensed data
- iii) Processing
- iv) Cloud Storage
- v) Security
- vi) Low power consumption
- vii) Scalability

IOT Ecosystem:

- An IOT Ecosystem is the complete system that includes everything to design IOT device that work together like Hardware, Software, Network, Users

Parts of IOT ecosystem:

- i) Device / sensors: It collects data from the environment.
- ii) Network: Sends data to the cloud on application
- iii) Data processing: It processes the data and convert it into useful information.
- iv) Cloud: It sends the data to the cloud to store safely.
- v) Applications: user see and control IOT devices
- vi) User: The person who uses the IOT system.

IOT decision framework:-

- An IOT decision framework is a step by step guide to help people to plan and build IOT systems by making smart choices.
- It helps them to decide:
 - i) what to build
 - ii) Who will use it
 - iii) What data it will collect

Areas of IOT decision framework

- i) User experience: It describes how users interact with it.
- ii) Data: what data will be collected from the IOT environment, how it will store and how to analyze it.
- iii) Business: It describes which problem will be solved.
- iv) Security: It is used to implement out the data and device will be protected.
- v) Operations: How will the system be managed after deployment like device updates, maintenance.
- vi) Devices: It shows which hardware devices are used like camera, micro controller, wifi.

IOT Architecture

Application Layer

Data processing Layer/
Intermediate layer

Network Layer

Perception Layer/
Sensor Layer

→ IOT architecture is the structure or design of how IOT devices (components) work together from sensing the environment to sending data, processing the data and finally showing the results to the user.

It is divided into four layers:

i) Perception Layer

→ It is also known as Sensor Layer which is responsible for sensing the data or collecting the information from the environment.

Devices used: i) Sensors ii) Actuators iii) Identification

i) Sensors: Temperature sensor, Humidity sensors, GPS

ii) Actuators: Motors, LED

iii) Identification: RF-ID, NFC

(II) Network layer

- This layer transfers data from sensing devices to the cloud using internet or communication network.

Technology:

- i) wifi
- ii) Ethernet
- iii) 4G, 5G

Protocols used to transfer data.

- i) HTTPS
- ii) IPV4, IPV6
- iii) TCP/IP

(III) Data processing layer / intermediate layer / middleware layer.

- This layer stores the data, processes and analyzes the data. It also makes the decision based on the data.

Technology used

- i) Cloud platforms: AWS, Azure, Google cloud
- ii) Databases: MongoDB, MySQL
- iii) AI & ML: for analyzing and processing data
- iv) Decision Engine: if-then, etc.

IV) Application layer:

- This layer displays the information to the user and allows them to control the IoT device.
- Examples: Mobile applications, web applications.

MicrocontrollerMicroprocessor

- | | |
|---|--|
| i) It is a small computer on a single chip with CPU, input, output ports and memory | - It is a chip that contains only CPU, it needs external memory and peripheral devices |
| ii) Designed to control specific device or task | Designed to process data and run complex applications. |
| iii) CPU + RAM + ROM + Input-output ports etc. | only CPU, RAM, ROM
Input-output ports are external. |
| iv) It is cheap as compared to microprocessor | It is more expensive. |
| v) It is small and compact | It is normally larger |
| vi) It require low power consumption | It require high power to operate. |
| vii) It is generally slower than microprocessor (usually Mega Hertz Range) | It is much faster (Giga Hertz Range) |
| viii) Example: Arduino, ESP 32, | Example: Intel i5, AMD Ryzen. |

Arduino UNO:

→ Arduino UNO is a micro controller board based on Atmega328P microcontroller chip. It is designed to make it easy for beginners and professionals to create electronic project and IoT devices by allowing them to connect sensors actuators and electronic components.

Different pins in Arduino UNO

① Power pins (left side area near the USB port)

→ This hub powers the sockets used to provide power supply to our code (3.3V pin, 5V pin, Ground pin (e), VIN). It is also used to power external battery or adapter.

② Analog pins (located on the left side of the board)

- A0, A1, A2, A3, A4, A5

- It reads analog signal like temperature, humidity, light intensity and convert them into digital numbers using ADC (Analog to digital converter) inside the Atmega328P.

③ Digital Pins (D0-D13)

- It is located on the right side of the board D0-D13.
Function:

- It reads and sense data in two modes.

Input mode - Read signal (on/off)

output mode - Sends digital signal

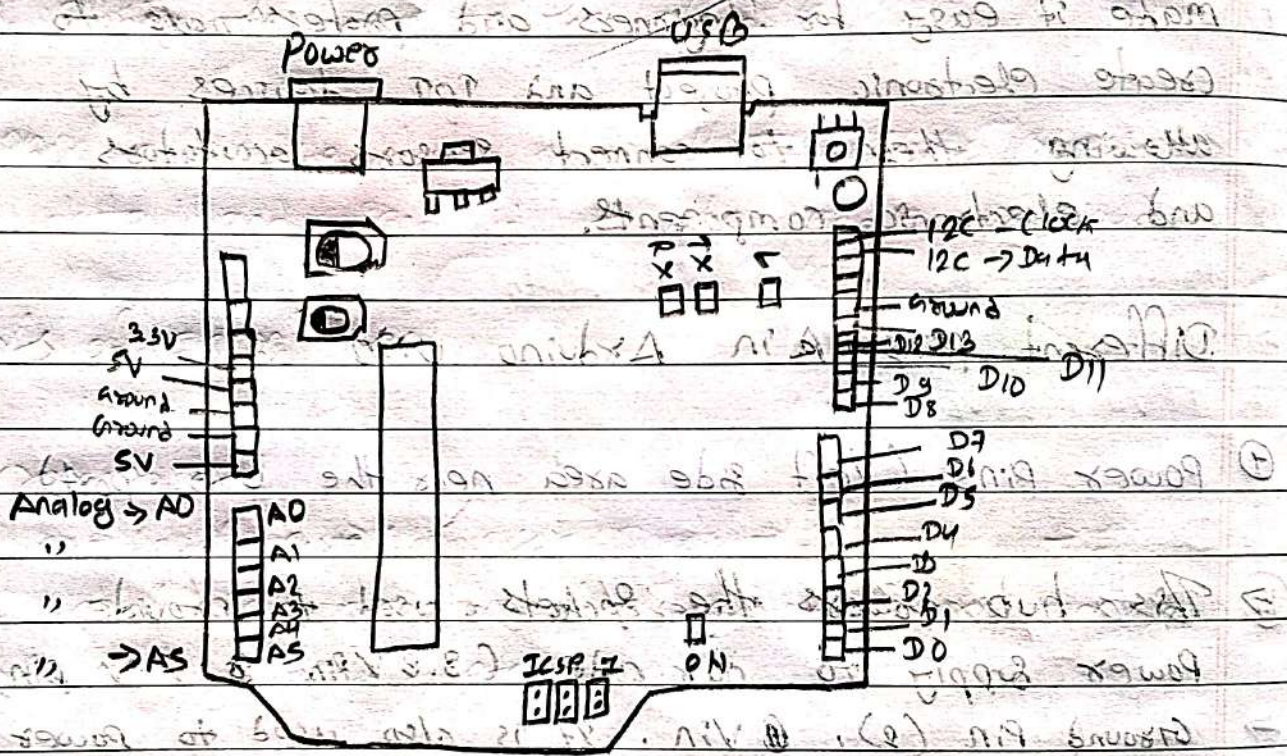
Tx = Transmission data
 Rx = Receives data
 ADC = Analog to digital converter

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⑩ PWM Pins

- It stands for pulse width modulation pin that can output variable voltage



D3, D5, D6, D9, D10, D11 : ~~D3~~ Analog output generation

(Analog pins are used for analog output generation)

ESP32 is a micro controller with build-in wifi and Bluetooth.

we can connect it through internet for sharing a data and storing the data to the cloud.

ESP32 reads the data from the sensor, process the data and send it to the mobile or cloud.

Parts of ESP32 are:-

i) Microcontroller

→ It has dual core processor

It controls all operations like reading sensors, turning port, sending signals etc.

ii) Memory

- It contains SRAM, Flash memory

-

iii) Wi-Fi Bluetooth

- This board has built-in wifi and bluetooth connectivity. This is the reason of popularity of ESP32

iv) GPIO pins

- It stands for General purpose input output pins, which is used for taking input and generating output. It is used for digital to Analog converter and Analog to digital converter.

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	Arduino	Node MCU 8266
Type	Microcontroller	Microcontroller
Speed	16 MHz	80-160 MHz
Memory	4Kb Flash	256 4mb
WiFi	No	Yes
Bluetooth	No	No
Ports	Analog $\rightarrow 6$, Digital $\rightarrow 14$	Digital $\rightarrow 1$, Analog $\rightarrow 1$

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	ESP82	Raspberry-Pi
Microcontroller	Microcontroller	Microprocessor
Speed	160-240 MHz	1.5 GHz
Memory	4mb	1-4 GB
WiFi	Yes	Yes
Bluetooth	Yes	Yes
Ports	GPIO $\rightarrow 36$	GPIO $\rightarrow 40$

Node MCU 8266

20923

→ It is a microcontroller board which is used to design IoT devices which can sense the data with the help of sensors and sends the data to the cloud with the help of wi-fi.

20923

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Pin board of Node MCU:

20923

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- i) It consists of a microcontroller chip which is ESP 8266
- ii) Flash memory: It consists 4 mb of flash memory.
- iii) USB port: This board contains one USB port to upload the program
- iv) USB to serial converter chip: This board has one USB serial chip converter that converts the USB data to serial
- v) Digital Analog pin: This board consists of 11 digital pins and 1 analog pin

GPIO = General Purpose Input Output Pins

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Raspberry Pi

- Raspberry Pi is a microprocessor which is small size, low cost, ~~that~~ credit card sized computer.
- It is used to run operating system like Linux, Windows IoT etc.
- It has multiple USB ports, 1 HDMI port, Audio Jack, ethernet port, 5V power input pin, forty GPIO.

Working of Raspberry Pi:

Bootling Process

Input

Processing

Output

Network connectivity

GPIO pins (External hardware access): Sensors

Arduino Programming Basics:

void setup(): This is the first method that runs when the Arduino is powered on.

pinMode(Pin, Mode): This method is used to set the mode of the pin.

digitalWrite(Pin, value): This method is used to write a digital value to the pin.

void loop(): This method is used to write the main logic of the program.

digitalRead(Pin): This method is used to read a digital value from the pin.

analogWrite(Pin, value): This method is used to write an analog value to the pin.

Void Setup

- It is a method in which we set the mode of the pin.
- This method runs only once when we run the code.

Void Loop

- This method runs continuously, all the logics are written inside this method.

Steps to create a smart Arduino device :-

- i) Firstly we create the circuit designing
- ii) In the coding part we set the pin mode of all the pins that are required in our project.
- iii) After setting the pin ^{mode} ~~board~~ we write the backend logic as per our requirement.

Note: There is a build-in LED which is used for testing purpose located at D13 pin.

Pin mode()

- This method sets whether a pin is input or output. ~~we~~ it accepts two parameters first is pin and second is mode.

Example: `pinMode(13, output)`

This means pin number 13 behaves as output pin. ~~8 = 13~~

digitalWrite()

- This is used to turn the pin ON. It also (have) two parameters first is pin and second is value.

Arduino coding is case sensitive?

1) ~~Analog~~ analogRead():

- This is used for analog pins to read the values from sensors

2) AnalogWrite():

- This is used to write - PWM (values whose range is from 0 to 255)

Tripple led setup in ARDUINO Board

Code -

```
int T1 = 2;
```

```
int T2 = 8;
```

```
int T3 = 10;
```

```
int T4 = 12;
```

```
void setup()
```

```
{
```

```
  pinMode(T1, OUTPUT);
```

```
  pinMode(T2, OUTPUT);
```

```
  pinMode(T3, OUTPUT);
```

```
  pinMode(T4, OUTPUT);
```

```
}
```

```
void loop()
```

```
{
```

```
  digitalWrite(T1, HIGH);
```

```
  delay(1000);
```

```
  digitalWrite(T2, HIGH);
```

```
  delay(1000);
```

```
  digitalWrite(T3, HIGH);
```

```
  delay(1000);
```

```
  digitalWrite(T4, HIGH);
```

```
  delay(1000);
```

```
}
```

Sensors:

- They are the devices which collect data from the environment.
- It detects physical changes like temperature, light, sounds, motion etc from the environment and convert it into electric signal that can be understood by the electronic devices.

Working principle of sensors

I) Input:

- It takes input from the environment changes
eg: temperature rises, light shines.

II) Sensing data:

- The core part of sensor react to this change.
eg: Photo-Diode: It generates current when light hits it.

III) Conversion:

- At this stage, the reaction is converted into electric signal.

IV) Output:

- That electric signal is send to microcontroller for processing.

Actuators:-

- It is an device that takes electric signal from the system and convert it into physical actions such as movement, temperature maintain, sound or light controlling.

Working Principle of Actuators

- I) Input: A microcontroller sends a small signal to a like on/off for a specific device

- II) Energy conversion: The actuator converts this electronic signal into energy.

Eg: moving of fan (converts electric energy into wind), (motor).

- III) Output: The actuator changes the real world state like a motor rotates the wheel, a loud speaker produces sound or a heater increases temperature.

Difference between Sensors and Actuators:

Sensors	Actuators
i) It senses data from the environment.	It takes electric signals from the system like microcontroller.
ii) The function of sensor is to detect the signal from the environment and convert it into electric signal.	It produces physical output like rotation of motor after receiving the electric signal from the microcontroller.
iii) Sensors produce only electric signal as output.	Actuators show physical actions.
iv) Eg: photodiode, temperature sensor, etc.	Eg: motor, speakers, heater etc.

MEMS (Micro Electro Mechanical System)

- MEMS is a small mechanical and electrical machine on a chip that consists of sensors and actuators.
- It is a combination of electrical and mechanical part.
- Its size is very small such that its moving part is not visible by normal eyes.

Why we need MEMS

- 1) Size: Its size is very small like a grain of a rice.
- 2) Power consumption: It consumes very less power that help for long lasting battery.
- 3) Fast response: All MEMS devices response very quickly like air bag of car.
- 4) Cheap: It is very cost efficient as compared to using normal sensors and actuators.

Uses of MEMS:

- 1) Accelerometer: MEMS accelerometers are used in phones that works when we rotate our mobile.
eg: 1) Portrait screen rotates into Landscape mode.
2) It counts our steps which are used in smartwatches.
3) In cars it is responsible to open the air bag when the car hits an object.
- 2) Gyroscope: It helps drone to balance in air.
- 3) Proximity sensors: It help to detect the object near by it. Like in mobile phones.

4) MEMS pressure sensor: It is used to measure the pressure like ~~in~~ⁱⁿ blood pressure machine, to measure the pressure of wheels of cars.

5) MEMS mic: MEMS is also used to design that are used in ear buds, mobilephones which contains a drum that senses the sound frequency.

MEMS can also be used as Actuators.

1) Ink-jet printer: Nozzles. This is used in printers that pushes tiny droplets of ink on the paper.

2) camera lense actuator: it is used to capture the clear image even when the camera is moving.

- Raspberry Pi Setup
- Raspberry Pi Imager
- Raspberry Pi OS
- SSH (Secure Shell) command line
- Teamviewer (Visuals)

Raspberry pi setup:-

- To setup Raspberry pi we need Raspberry pi board, SD card (minimum 16 GB), card reader. SD card act as Hard disk for Raspberry pi.
- we need to download Raspberry pi imager software which is useful to flash raspberry OS on the SD card.
- After opening the raspberry pi imager software we need to select the OS that we want to flash on SD card, then we select the storage device where we want to flash.
- After selecting the storage device we need to click on the setting and write the user name and password for raspberry - pi OS. Activate the host name and SSH so that we can connect it through another computer.
- We need to setup the wifi name and password through which we want to connect our raspberry - pi board.
- After setting all these fields click on submit button. Now we can use the SD card in raspberry - pi board and connect monitor, keyboard and mouse to work with raspberry - pi.

Communication Protocol used in IOT Devices.

communication means sharing of informations between two persons devices etc.

Protocol

- Protocols are the set of rules that establish a successful communication between two devices.

Why we need communication protocol

Different IOT devices are made of with different hardware configuration that have different encoding methods, different packet size

Example:- Different companies make smart bulbs that have different hardware configuration, different format packets and different encoding methods. If there is no communication rule then we cannot control the same smart bulb that are manufactured from different companies

→ This communication Rule is called protocol

When we use IOT protocols.

1) Device to Device

→ It is used to when we want to send our information from one device to another device without using Internet or any gateway

Example: if we connect our mobile phone with smartwatch then we don't need an internet connection and any gateway to transfer our data.

II) Device to Gateway

- This communication is used when we want to send our data from a device to gateway like a router.

Example: if a smart bulb sends the data to the router for further processing, then this is called device to gateway.

III) Device to Router in cloud :-

→ In this communication protocol, the IoT device cannot send data directly to cloud so it sends the data to the router and the router retransmits the data to the cloud.

Example: if a camera sends the data to router and it retransfers the data to cloud.

Wireless Communication:-

→ wireless communication means communicating with different devices without cables.

In wireless communication we are using radio wave, infrared, satellite signals etc.

Why we need wireless communication in IoT devices

→ IoT means (Internet of things) and there are billions of IoT devices which are very small in size. It is very complex and expensive to connect IoT device through wires for communication.

There are 2 types of wireless communication devices based on range communication ranges.

① Short-range communication:-

→ It covers small distance from few meters to 100 meters.
Example:

- i) Bluetooth
- ii) WiFi
- iii) Zig-bee

② Long-range communication:-

→ It covers large distance in kilometers.

Example:

- i) Cellular network
- ii) LP-WAN
- iii)

1) Bluetooth:-

- It is a wireless communication device which uses radio wave for transferring of data. The frequency of Bluetooth is 2.4 GHz and it covers the range of 10 meters to 100 meters depending upon the version.

- It's power consumption is very low and it is used in smartwatch, mobile phone, earphones, etc.

Pico net

- A Pico net is a small Bluetooth network that is made up of one master and up to seven slaves.
- Slave can communicate with master only, if one slave wants to send information to another slave then it needs to send the data to the master and the master and the master send it to another slave.

Master:

The device that controls the communication, it decides when and how data is sent. These are the devices that follow master instruction.

Wi-Fi

- Wi-Fi means accessing internet or sharing information without wires. It uses radio wave by the help of which we can connect our Wi-Fi enabled devices through internet.

Working of Wi-Fi

- Internet comes to our home with the help of cables and connected to wifi routers.
- Wi-Fi routers receives the signals from the wire and convert it into radio wave signal. Our Wi-Fi enabled devices captures the radio wave signals and connects to the internet.
- Whenever we search anything then the request goes to the router and these request further sent to the internet service providers.

Application of Wi-Fi

- By the help of Wi-Fi, we connect our Wi-Fi enabled devices to internet, or share our information from one device to another device.
- we can use Wi-Fi as hotspot also.
- we can use Wi-Fi for VoWi-Fi to increase the quality of sound, when we are talking with our friends through internet.

uses:-

- Wi-Fi can be used as WLAN, we can construct simple wireless connection from one point to another.

Wireless Accesspoints

- It is a central point that produces or spread wireless signal through the help of ~~sent~~ which we are accessing internet.

Zig-bee

- It is a wireless communication technology used for short distance and lower power consumption.

It help smart devices to communicate with each other without wires.

Example

- I) Light automatically turn on when it detects any movement.
- II) Fan starts when the temperature increases.

Why we need zig-bee in IOT.

- For IOT devices we need to send small data temperature value so we use zig-bee for transferring of data. It consumes less power and gives long battery time.

- It communicates through another devices for small area.

Example

A temperature sensor needs to send small information so it don't needs wi-fi for high speed internet.

Components of zig-bee.

- I) coordinator
- II) Router
- III) End devices

Coordinator:-

- It manages and communicates with routers and end devices.

Router

- It's used for passing the data between devices.

End-Devices

- It's the devices that is being controlled.

Note:-

There is only one zig-bee coordinator in the network and may be multiple routers and end-devices.

Zig-bee Architecture Layers

- i) Physical Layer:- This layer is responsible to send & receive data as radio waves.
- ii) MAC Layer:- It is responsible to control the devices that means which device ^{data from} sends the data.
- iii) Network Layer:- It finds the best path to send the data.
- iv) Security Layer:- This layer is responsible for encrypting our data to provide security.
- v) Application Interface Layer:- This layer helps the application to talk to the network. e.g:- Zig-bee coordinator receives data from temperature sensor.
- vi) Application Layer:- This contains the actual device logic.
 Examples: Bulb application understands on/off commands.

wireless smart home sensors used actuators working

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6LOWPAN

→ It stands for IPv6 over low power wireless personal area network. It is used to connect our devices through the internet.

Basically 6LOWPAN is not a device it is a protocol that have set of standards or rules that helps our IOT devices to connect or send the data through the internet.

This is basically used in small IOT devices and this is a networking protocol that consumes very low power as compared to Wi-Fi or Bluetooth.

wireless

Long Range Communication devices

- If our IOT devices needs to send the data for a long distance like in kilometers then we do not consider with Bluetooth, Zigbee or 6LOWPAN

For sending our data over large distance then we have mainly two range of IOT options

- 1) Cellular IOT
- 2) LPWAN

1b) Cellular IoT.

- It uses existing mobile network towers to connect IoT devices to send the data. It ranges in between 2 to 40 km.

Example:

Vehicle tracking system has a small GPS and narrow bandwidth IoT device that measures GPS location, speed.

- Satellite sends signal and GPS device receives 4 satellite signal at a certain time where each satellite signal sends their position and the time when the signal is sent.

By the help of satellite position and signal sending time, the GPS calculates its exact position. After calculating the exact position it sends the data through the help of network to the cloud.

Industrial Internet 4.0

Industry Internet 4.0 means smart factories where machines, sensors, robots and computers are all connected through internet. It is also called Industry 4.0.

Components of industry 4.0

- i) IoT sensors : collect data from machine, product or environment. (temp, vibration etc)
- ii) Smart machines: These machines collect information from sensors and make small decisions without the help of human being
ex: Packaging machine, smart fan in factories
- iii) Cloud storage: All machine data goes to cloud where the processing occurs. From the cloud storage any authorized user can access the info about the machines ex: Remote inventory management.
- iv) Data Analytics: It process the data and find the pattern that help to predict machine failure, improve efficiency and plan production.
- v) Automation : Automation means performing different tasks without the involvement of human being. Example: moves goods from one place to another with the help of robot.

① What are the short-range communication devices?
Compare any three.

→ Short-range communication devices are wireless technologies that transfer data over a short distance (typically 1-100 meters) with low power consumption.

They are common in IoT because they allow quick, local, energy-efficient connectivity between sensors, wearables, and gateways.

Examples: Bluetooth, Wi-Fi, ZigBee, NFC

Comparison of any three

Feature	Bluetooth	Wi-Fi	ZigBee
Range	10-30m	50+ m	10-100m
Data Rate	1-3 mbps	100-1000 mbps	250 kbps
Power use	very low	high	very low
Network type	Point-to-point	star	mesh
Use case	wearables, Audio	internet access	Smart Homes

② Describe in detail the working of image and motion sensors

→ Image sensor working

- An image sensor converts light into digital signals to create images.

Explanation:

- made of millions of photodiodes arranged in a grid (pixels).
- when light hits the photodiode, it generates electrical charge.
- The charge is converted via ADC (analog to digital converter).
- Processor forms a usable digital image.

Types: CMOS (common in IoT - low power)

CCD (higher quality but power hungry)

use case: smart cameras, drones, surveillance

Motion sensor working (PIR sensors)

- motion sensors detect movement of objects or humans.

Explanation:

- PIR → Passive infrared sensor
- Detect changes in IR radiation from warm bodies.
- When a person moves, IR pattern shifts → sensor outputs HIGH.

use case:

Smart lighting, security alarms, automatic doors.

③ what is the full form and use of 6LoWPAN?

→ Full form

IPv6 over Low-power wireless Personal Area networks

Definition & use

- A Networking protocol that lets tiny, low-energy IoT devices use IPv6 over IEEE 802.15.4

It enables:

- Large IP address space
- Low power
- Direct internet routing

use case

Smart homes, industrial sensors, mesh IoT networks

④ what are long-range wireless communication types?

→ Long-range communication technologies allow data transfer across kilometers with low bandwidth and low power.

Examples:

LoRaWAN - Long range, Super low power

Cellular IoT - From telecom towers

Satellite IoT - Remote areas (Iridium, Starlink IoT)

Sigfox - ultra-narrowband, long-range

LTE-M - Higher speed cellular IoT

use case:-

Smart agriculture, smart meters, logistic tracking.

⑤ Mention any three benefits of Industrial Internet 4.0.

→ Industrial Internet 4.0 means factories where machines, sensors, Robots and computers are all connected through Internet. It is also called Industrial 4.0.

Benefits:

- i) Automation & Efficiency
Machines communicate & make decisions → faster production
- ii) Predictive maintenance
Sensors detect wear → reduces downtime
- iii) Better Data Analysis
Real-time insights improve quality and output
- iv) Safety measures
- v) Low production cost

⑥ Define Raspberry Pi and mention its basic components.

→ Raspberry Pi is a small, cheap, credit-card sized computer developed by Raspberry Pi Foundation (UK) in 2012.

It works like a normal PC but is smaller and affordable, making it popular for IoT, robotics and electronic projects.

Components

- Processor (CPU + GPU)
- RAM : Temporary memory (1-8 GB)
- USB port : for keyboard, mouse, camera, pendrive
- HDMI port : Monitor
- Ethernet port : Internet wired
- WiFi & Bluetooth : wireless
- Micro SD card slot : for OS and storage
- GPIO Pins (40) : for sensors and actuators
- Power input (5V) : power to the chip

⑦ What precautions should be taken while powering Raspberry Pi?

→ Precautions:

- i) Always use official 5V-3A adapter.
- ii) Avoid cheaper charger / adapter.
- iii) Use a good quality class SD card.
- iv) Always shut down properly before removing power.
- v) Never unplug during write operation.

⑧ Compare Bluetooth, Wi-Fi, Zigbee and 6LoWPAN.

→ Other are done in Q1.

6 LoWPAN

Features of 6LoWPAN

Range: 10-100m

Data Rate

Similar to Zigbee

Power use

Very low

Network type

mesh

use case

Low-power IPv6 network.

⑨ Write short note on accelerometer.

→ An accelerometer measures acceleration forces on an object.
Explanation

- Detects changes in velocity or orientation.
- Works using tiny MEMS structures that bend under motion → generate electric output.

uses

- Phones (screen rotation)
- Wearables (steps counting)
- Drones (flight stability)

⑩ Describe in detail the working of smart Home IoT system with example.

→ A smart home system uses interconnected IoT devices to automate home functions like lighting, security, and appliance control.

Explanation:

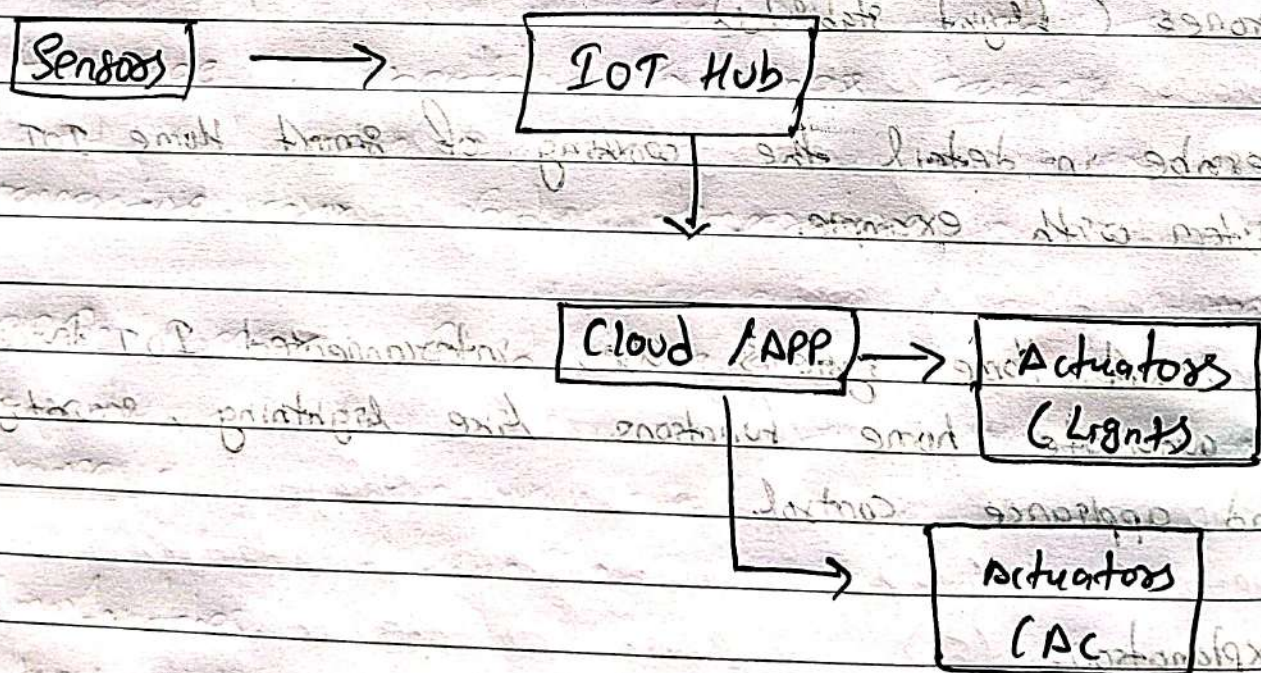
- Devices connect via Wi-Fi / ZigBee / Bluetooth.
- Sensors collect data (light, motion, temperature).
- Data goes to IoT hub or cloud platform.
- Automation rules trigger actions.

Example :

Smart Lighting system

- i) Motion sensor detects entry.
- ii) Light level sensor checks brightness.
- iii) If dark $\rightarrow$ lights turn ON automatically.
- iv) user can control light via smartphone app.

Block Diagram



(11) Explain how WiFi is used in IoT devices

→ WiFi is a high-speed wireless communication protocol used to connect IoT devices to local networks and the internet.

Explanation:

- IoT devices with WiFi modules (ESP8266, ESP32, Raspberry Pi) connect to routers.
- Router → cloud → mobile App → user controls device.
- Good for high-data applications like cameras.

Examples:

- smart cameras streaming videos
- smart assistants (Alexa, Google Nest)
- smart appliances (washing machines, A/C)

⑫ Explain how sensors help in wearables.

→ Wearables are IoT devices worn on the body, using sensors to monitor movement, health and environment.

How sensors help:

- Accelerometer: steps, activity, sleep tracking
- Heart-rate sensor: pulse monitoring
- Gyroscope: orientation for fitness tracking
- Temperature sensor: Body temperature
- SPO₂ sensor: Blood Oxygen.

Examples:

- Smartwatches
- Fitness bands
- Smart shoes

Q13) write the advantages of using Bluetooth in smart devices

- (Advantages)

- i) Super low power $\rightarrow$ perfect for wearables
- ii) Easy pairing $\rightarrow$ phones, headphones, sensors
- iii) Low cost $\rightarrow$ cheaper than wifi modules
- iv) stable short range communication & Reliable data sync.
- v) It uses a "sleep and wake up" system.

(14) Describe the complete setup process of Raspberry Pi with diagram

- i) Download OS: Get Raspberry Pi OS from official websites.
- ii) Flash OS: Use Raspberry Pi Imager to write the OS onto a micro SD card.
- iii) Insert SD card: Place it into the Pi's card slot.
- iv) Connect Devices: Keyboard, mouse, monitor and other devices.
- v) Power ON: Use a 5V-3A adaptor to start the Pi.
- vi) First boot setup: Choose language, user id, password, host name, WiFi, with connections and update software.
- vii) Ready to use: Now Pi works like a mini-computer where you can browse, code in Python and connect devices.

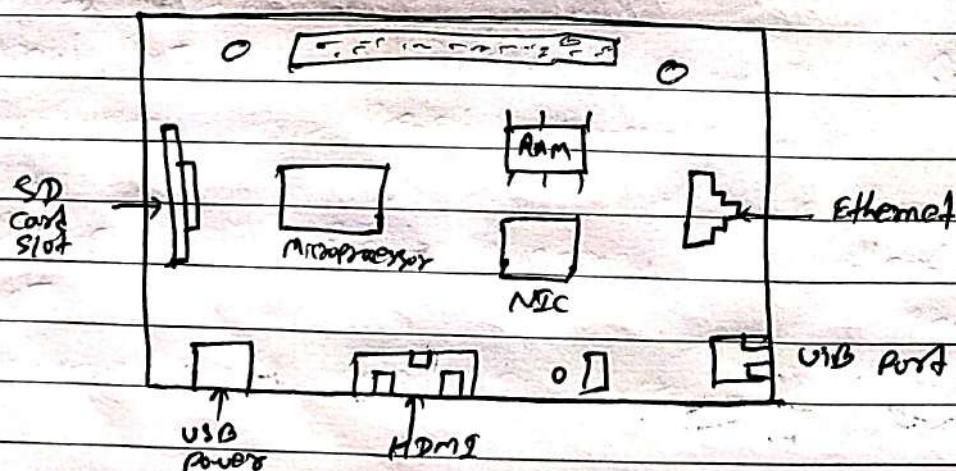


Fig: Raspberry - Pi

⑮ Compare zigbee and wifi in IoT usage

→ Feature	Zigbee	Wi Fi
Range	10-100m	50+ m
Power use	Very low	High
Speed	250 kbps	100+ Mbps
Network type	Mesh	Star
Best use	Smart home	Internet-connected devices
Battery life	Long	Short

Conclusion

- ZigBee → ideal for sensors & smart homes
- Wi Fi → better for high-speed devices like cameras.

⑩ Create a comparative table for all 7 IoT applications discussed

→ Assuming the common 7 applications: smart Home, smart Grid, wearables, smart city, smart Retail, Industrial IoT, Healthcare IoT.

IoT Application	Key features	Example Device
i) smart home	Automation, remote control	smart light, AC, smart lock
ii) smart Grid	Efficient energy Distribution	smart meter
iii) wearables	Health / activity monitoring	smartwatch
iv) smart city	Traffic, parking, Pollution,	smart streetlights
v) smart retail	Automated billing, inventory	smart shelves
vi) Industrial IoT	Automation, Predictive maintenance	Robots
vii) Healthcare IoT	Remote monitoring	Heart-rate sensor

(17) types of wireless communication used in IoT are

- i) short Range
 - Bluetooth (smartwatch)
 - ZigBee (smart bulbs)
 - NFC (payments)
- ii) medium - Range
 - Wi-Fi (smart camera)
- iii) Long - Range
 - LoRaWAN (agriculture sensors)
 - NB-IoT (smart meters)
 - Sigfox (asset tracking)
- iv) Cellular
 - 4G, 5G, IoT 5G (Car, GPS trackers)

- (18) Feature
- | | | |
|------------|------------------------|------------------|
| Speed | Low (250 kbps) | High (1-3 mbps) |
| Energy use | very low | low |
| Use | IPV6 sensors, networks | wearables, audio |

(19) Define smart home and list two devices used

- A smart home is an IoT-enabled living space where devices automatically operate using sensors, internet connectivity and automation rules

Devices:- 1) Smart lights

2) Smart thermostat

② Methods to Access the Raspberry Pi remotely

I) SSH

→ SSH (Secure Shell) is a protocol that allows remote access to Raspberry Pi terminal (Command line) from another computer or mobile device.

How to use it

- I) Enable SSH in Raspberry Pi settings
- II) Find Pi's IP address (e.g. 192.168.10.1)
- III) On another device open an SSH client (e.g. terminal)
- IV) Type command `ssh pi@192.168.10.1` and enter the password
- V) Now you can control the Pi remotely without needing a separate monitor or keyboard

II) Team Viewer

This software gives full desktop access with GUI controls and we can transfer files from one computer to another easily.

Steps to use it

- I) Install teamviewer host on Raspberry Pi
- II) Install teamviewer on laptop or mobile
- III) Create a login to the same teamviewer on both devices
- IV) On laptop/mobile enter teamviewer ID of Raspberry Pi.
- V) The full Raspberry Pi desktop will appear, allowing you to control it as if you're sitting in front of it.

Q1) What is Smart Retail? Give two uses.

→ Smart retail is the application of IoT technology in shopping environments to automate processes, enhance customer experience, and optimize store operations.

Uses:

1) Smart Shelves

- use weight sensor + RFID to detect stock levels automatically.

2) Automated Billing / Chashiers-less checkout

- AI + sensors track items and charge customers automatically (like Amazon Go store).

Q2) What is a proximity sensor used for?

→ A Proximity sensor detects the presence of an object without physical contact.

Use:

- used to sense distance or detect objects approaching a device.

Example:

- Phones turning off the screen.

- Automatic doors

- Robotics obstacle detection.

② List 3 types of IoT boards and describe in brief.

① Arduino Uno R4

→ It is a microcontroller board based on ATmega 328P chip.

- No OS system in it. External wifi or bluetooth connectivity.

Speed: 16 MHz, memory 4 kb Flash

Pins: Analog 6, Digital 14

Usage: sensor based projects, automation, simple IoT prototypes.

② Raspberry Pi

→ Microcontroller board, can have basic OS like Linux, USB port, HDMI, audio, ethernet.

Speed = 1.5 GHz, memory = 1-4 GB, GPIO = 40

Usage: edge computing, home automation, AI edge, Run python, Docker.

③ ESP 32

- Micro controller board with built in wifi, bluetooth, read sensor data, process it and send to mobile or cloud.

Speed = 160 - 240 MHz

memory = 4 MB

GPIO → 36

②4 Explain GPS, GLONASS, IRNSS, and Galileo with their differences.

- | | |
|------------------------------|-----------------------------------|
| ⇒ GPS (USA) | GLONASS (Russia) |
| • 31 satellites | • 24 satellites |
| • Global coverage | • Global coverage (maintains) |
| • widely used for navigation | • slightly better in cold regions |
| • Accuracy 5-10m | • Accuracy 5-7m |

- | | |
|--|-----------------------------------|
| IRNSS / NavIC (India) | Galileo (EU) |
| • 7 satellites | • 30 satellites planned |
| • Regional coverage (India + 1500 km radius) | • High precision civilian service |
| • Highly accurate locally | • very accurate in urban |
| • Accuracy ~ 3m (India) | • Accuracy < 1m |

②5 Explain the process of gyroscope calibration and noise filtering

→ Gyroscope calibration ensures accurate measurement of angular motion by correcting errors like drift and bias.

Calibration steps

- 1) Keep sensor stationary (Raw data)
- 2) Find bias (offset value)
- 3) Subtract bias (subsequent reading)
- 4) Repeat periodically

(26) write detailed notes on sensing actions with real-life examples.

→ Sensing actions refers to how sensors collect data from the environment and trigger responses in IoT systems.

Explanation:

Sensors detect physical changes (light, sound, motion, temperature) and convert them into electrical signals → processed by microcontroller → actions occur.

Real-life examples

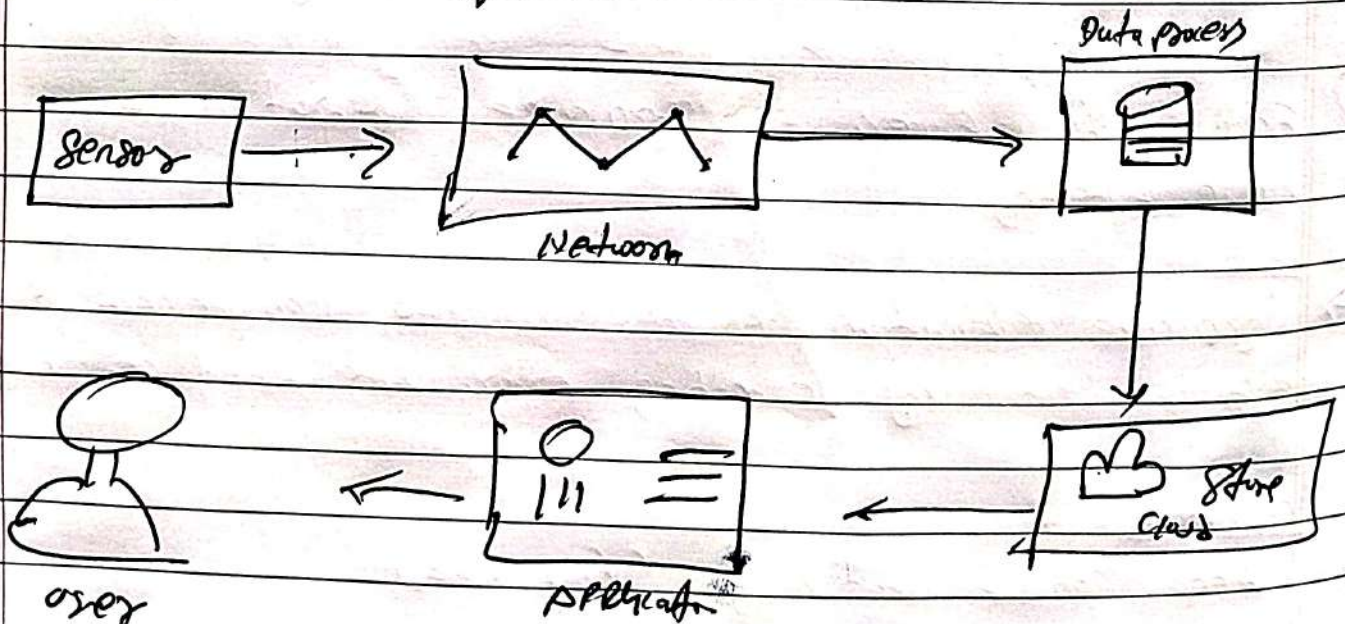
- Smart-AC: Temperature sensor detects high temp → AC turns ON.
- Streetlight: Light sensor detects darkness → lights activate.
- Fitness bands: Accelerometer tracks movement → steps counted.

Components of IoT solution architecture or ecosystem of IoT

→ The IoT solution architecture or ecosystem is the complete system needed for IoT devices to work together including hardware, software and networks.

Components:-

- Devices / sensors : collect data from environment
- Networks : encrypt and transfer data.
- Data processing : convert into usable form
- cloud storage : store data safely
- Application / App : all to see usable dashboard or control
- user : uses the system



IOT architecture or working of IOT

→ IOT devices work together from sensing the data from environment, collect it to cloud through internet and perform basic operations at the application part and gives result to user. It has 4 layers which are:

i) Sensor layer (Presentation layer)

- Also called P.L, It senses data

ii) Network layer

→ data from sensing device to cloud

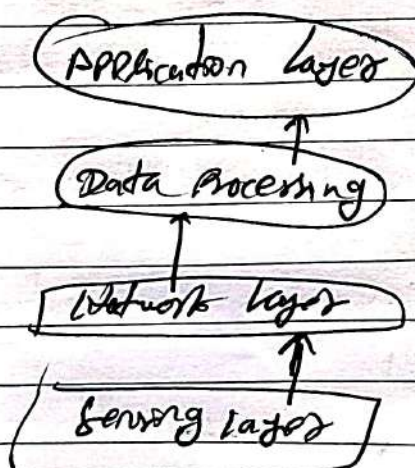
Device: 4G, 5G, Protocol: HTTP, IPV4, TCP

iii) Data processing layer (Intermediate layer)

→ Store data, process and analyze to make decisions per server, DB & ML, cloud platforms.

iv) Application layer

- This layer displays the info to the user and allows them to control the IOT devices ex: mobile apps, web applications



Total No. of Pages: 02

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ESE (Odd Sem) Dec 2025
B.Tech in CSE (AI & ML)

Semester-III

CSAIML306A: Open Elective-I (Internet of Things)

Time: 2:30 Hours

Max Marks: 50

Instructions to the candidate:

- Question Paper consists of 3 Parts (Part-A, Part-B & Part-C)
- Figures to the right indicate full marks.
- Draw neat sketches and diagrams wherever necessary.

PART- A

(MCQs Types)

Answer All Questions (1 x 10 = 10 Marks)

Q. N.	Questions	Mark	CO
1	Smart bulb is an example of: A) Mobile B) Laptop C) IoT Device D) Chalk	1	CO1
2	Which one is not an IoT device? A) Smart fridge B) Smartwatch C) Smart car D) Paper Notebook	1	CO1
3	What is SSH used for? A) Playing games B) Drawing	1	CO2

	C) Remote login D) Watching videos		
4	Which tool helps in accessing Raspberry Pi remotely? A) Paint B) SSH Client C) Wordpad D) Excel	1	CO2
5	Which protocol is best for low energy use? A) WiFi B) Ethernet C) Bluetooth Low Energy D) USB	1	CO3
6	ZigBee range is: A) 1 meter B) 10 meters C) 10-100 meters D) 1 km	1	CO3
7	Which of the following is an example of IoT in smart cities? A) Blackboard B) Smart street lights C) Manual fan D) Typewriter	1	CO4
8	Which is a key benefit of telemedicine? A) Travel long distances B) Delay in treatment C) Remote diagnosis D) Use of typewriters	1	CO4
9	What does an accelerometer measure? A) Speed B) Light C) Movement D) Noise	1	CO5
10	Indoor localization is helpful in:	1	CO5

	A) Flying B) Mapping C) Room tracking D) Weather		
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PART-B

Answer Any Four Questions- (5 x 4 = 20 Marks)

Q.N.	Questions	Marks	CO
<u>1</u>	Explain the role of sensors and actuators in IoT.	5	CO1
<u>2</u>	What precautions should be taken while powering Raspberry Pi?	5	CO2
<u>3</u>	What are long-range wireless communication types?	5	CO3
<u>4</u>	What is Smart Retail? Give two uses.	5	CO4
<u>5</u>	Give two advantages of Smart City IoT.	5	CO4
<u>6</u>	What is a proximity sensor used for?	5	CO5

PART- C

Answer Any Two Questions- (10 x 2 = 20 Marks)

Q.N.	Questions	Marks	CO
<u>1</u>	What are short-range communication devices? Compare any three.	10	CO3
<u>2</u>	Create a comparative table for all 7 IoT applications discussed.	10	CO4
<u>3</u>	Describe in detail the working of image and motion sensors.	10	CO5